

Literature Survey: MoE Serving, Expert Offloading & Speculative Prefetching

Positioning the AAEC v3 Serving Engine

1. Survey Scope & Methodology

This survey covers **42 papers** (2020–2026) organized into **7 categories** that collectively define the design space in which AAEC v3 operates. For each paper we extract: core idea, method, evaluation setup, key results, weaknesses, and stated future work. We then perform cross-paper gap analysis to identify where AAEC v3's contributions are truly novel.

2. Taxonomy of Categories

[Literature Survey Taxonomy: Foundational -> Offloading -> Speculative -> Sparsity -> Optimization -> Infrastructure -> Quantization]

3. Category C1: Foundational MoE Architectures

These papers define the MoE model designs that AAEC v3 serves. They are not competing systems — they are the *workloads*.

#	Paper	Year	Idea	Method	Key Results	Weakness (for Serving)
1	GShard (Lepikhin et al.)	2020	Scale MoE to 600B params	Top-2 gating + expert capacity factor + automatic sharding	BLEU +13.5 on MT	Expert capacity overflow drops tokens; no serving latency analysis
2	Switch Transformer (Fedus et al.)	2022	Simplify to Top-1 routing	Single-expert routing + auxiliary load-balancing loss	7× pre-training speedup over T5	Top-1 routing limits expert diversity; load imbalance persists
3	Mixtral 8×7B (Jiang et al.)	2024	Open-source sparse MoE LLM	8 experts per layer, Top-2 routing, 47B total / 13B active	Matches LLaMA-2-70B quality	Monolithic 47B weight footprint; no offloading-aware design
4	DeepSeek-V2 (DeepSeek)	2024	Fine-grained MoE + MLA	160 fine-grained experts + 2 shared experts per layer; MLA for KV compression	236B total / 21B active; matches dense 70B	160 experts ×48 layers = 7,680 expert matrices; massive offloading pressure
5	DeepSeek-V3 (DeepSeek)	2024	Auxiliary-loss-free MoE	Bias-based load balancing; MTP training; FP8 training	671B / 37B active; top-tier quality	256 routed experts per layer; extreme memory pressure for edge serving
6	Qwen3-30B-A3B (Alibaba)	2025	High-sparsity mobile MoE	128 experts per layer, Top-8 routing, 30B total / 3B active	Strong quality at 3B active params	Top-8 routing across 128 experts creates 6,144 possible expert calls per layer

7	LLaMA-MoE (Zhu et al.)	2024	Upcycle dense -> MoE	Partition FFN layers into experts + continual pre-training	Recovers 80-90% of dense performance	Inherits dense model's activation patterns; cache behavior unknown
8	Expert Choice Routing (Zhou et al.)	2022	Experts choose tokens	Experts select top-k tokens instead of tokens selecting experts	Perfect load balance; +2% quality	Not usable for autoregressive B=1 inference (requires full-batch token pool)

*[!NOTE] Key observation for AAEC v3: Modern MoE models (DeepSeek-V3, Qwen3) use 128–256 fine-grained experts with Top-8 routing. This means the working set per token is large (8 experts × 768 columns = 6,144 columns per layer), but each individual expert is small (9.44 MB for Qwen3). This fine granularity is what makes **column-level** caching viable — a contribution absent from all C1 papers.*

4. Category C2: Expert-Level Offloading & Caching

These are the **direct competitors** to AAEC v3. They all operate at expert-level granularity.

#	Paper	Year	Idea	Method	Eval Model / Dataset	Key Results	Weaknesses	Future Work
9	Mixtral Offloading (Eliseev & Mazur)	2023	LRU cache + speculative prefetch	Expert-level LRU cache on GPU; predict next-layer experts from current hidden state	Mixtral-8×7B / WikiText	2-3 tok/s on consumer GPU; ~93% cache hit with LRU-2	Expert-granularity only; LRU eviction has no lookahead; speculative loading from <i>previous layer</i> (not pre-attention)	Smaller expert groups; better prediction models
10	MoE-Infinity (Xue et al.)	2024	Activation tracing + temporal locality	Sequence-level expert activation trace; exploit temporal locality for prefetching	Mixtral-8×7B, DeepSeek / LM benchmarks	2.4× throughput over Mixtral-Offloading	Expert-level; no column slicing; requires activation trace history	Sub-expert offloading; cross-request locality
11	MoE-Lightning (Cao et al.)	2025	CPU-GPU-I/O pipelining	CGOPipe scheduler + paged expert weights	Mixtral-8×22B / throughput benchmarks	Up to 10× throughput on single GPU	Throughput-optimized (high batch), not latency-optimized (B=1)	Latency-oriented scheduling
12	CommitMoE (Luo et al.)	2025	Commit Router (fallback-free)	Router commits to 1 expert pre-attention; skips other 7	Mixtral-8×7B / quality + latency	1.68 ms stall but drops 7/8 experts (lossy)	Catastrophic quality loss (>87.5% expert content discarded); not suitable for production	Recovering dropped expert information
13	ADEPT (2026)	2026	Domain-aware preloading + locality prefetch	Two-stage: domain preloading during prefill, locality-aware prefetching during decode	DeepSeek-V2 / mixed benchmarks	Improved cache hit rates vs. LRU	Expert-level granularity; no sub-expert decomposition	Finer-grained expert management

14	DALI (2026)	2026	Workload-aware 0-1 optimization	Integer programming for CPU/GPU expert assignment + residual-based prefetching	DeepSeek-V3 / local PC	Dynamic assignment; low stall on PCIe	Still expert-level; IP solver overhead; no attention-window hiding	Multi-node extension
15	FineMoE (Wang et al.)	2026	Fine-grained offloading with semantic hints	Extract expert selection patterns + prompt semantics to guide caching	DeepSeek-V2 / LM benchmarks	Better cache hit than LRU/LFU	"Fine-grained" refers to prediction granularity, not sub-expert weight slicing	Cross-layer dependency modeling
16	EdgeMoE (Yi et al.)	2023	MoE on mobile devices	Storage hierarchy partitioning + expert-wise bitwidth adaptation + preloading	Mixtral on smartphone	10× speedup over naive offloading	Expert-level; mobile-only (Flash storage, not PCIe); mixed precision is lossy	Sub-expert slicing for mobile
17	MoEpic (2025)	2025	Adaptive expert split	Vertical expert splitting for improved cache hit	DeepSeek / throughput	Better transfer-computation overlap	Splits are coarse (halves/quarters), not column-level	Dynamic splitting ratios
18	ExpertFlow (2025)	2025	RL-guided expert caching	Reinforcement learning for expert selection + cache management	MoE models / benchmark	Dynamic expert placement	RL overhead; expert-level granularity	Cheaper prediction models
19	ExpertCache (2025)	2025	RL-guided 2-phase caching	Pre-loading controller + runtime controller via RL	MoE models	Reduced VRAM requirement	Expert-level; RL training cost	Integration with quantization
20	MoE-Gen (2025)	2025	Module-based batching	Accumulate tokens, batch-launch expert computation	DeepSeek-V3 / single GPU	High throughput on single GPU	Optimized for throughput, not latency; adds token accumulation delay	Latency-sensitive serving
21	MoE-Beyond (2025)	2025	Learning-based expert prediction	Lightweight transformer predicts multi-label expert activation	Edge devices	Cache hit: 17% -> 72%	Expert-level prediction; model-specific training needed	Transfer to new models
22	ProMoE (2025)	2025	Proactive caching	Uses intermediate results to predict next-layer experts	Mixtral / LM benchmarks	Eliminates passive cache misses	Expert-level; requires per-model calibration	Sub-expert caching
23	OD-MoE (2026)	2026	Cacheless distributed edge	Parallel expert loading + predictive forecasting; no cache	Edge devices <1GB VRAM	Inference on extreme edge	No cache = every expert loaded from storage; high bandwidth demand	Expert compression

*[!!IMPORTANT] **Critical gap across ALL C2 papers:** Every system in this category operates at **expert-level granularity** (9.44 MB per expert for Qwen3). None performs sub-expert column-level decomposition. This is the single most repeated limitation in the literature.*

5. Category C3: Speculative Prefetching & Prediction

#	Paper	Year	Idea	Method	Key Results	Weaknesses	Future Work
24	Pre-gated MoE (Hwang et al., Microsoft)	2024	Pre-gating function for offloading	Algorithm-system co-design: pre-gating predicts experts before routing	Enables cost-effective deployment	Prediction from previous layer's output (inter-layer), not pre-attention (intra-layer)	Intra-layer prediction
25	SpecMoE (2025)	2025	Self-assisted speculative decoding	Uses speculative decoding's draft stage to forecast expert usage	Accurate expert prediction without external draft model	Requires speculative decoding framework; not applicable to standard autoregressive serving	Integration with standard serving
26	MoE-SpeQ (2025)	2025	Speculative quantized decoding	Entropy-aware caching + amortization roofline model for optimal draft length	Improved prefetch efficiency with mixed precision	Requires external draft model; quantization introduces small quality loss	Lossless speculative drafting
27	AdapMoE (2024)	2024	Adaptive sensitivity-based gating	Dynamically adjusts number of active experts per layer based on token sensitivity	Reduced loading overhead	Lossy: reduces number of active experts (changes model semantics)	Lossless adaptive gating
28	HOBBIT (2025)	2025	Mixed precision expert offloading	Replaces cache-miss experts with low-precision versions at runtime	Reduced loading latency	Lossy: substitutes FP16 experts with INT4 at runtime	Lossless mixed-precision serving
29	SiDA-MoE (MLSys 2024)	2024	Sparsity-inspired data-aware serving	Parallel prediction + inference threads; data-aware expert placement	3.93× throughput, 72% latency reduction	~1% quality degradation (lossy); expert-level	Zero-quality-loss serving
30	ST-MoE (Prefetch)	2025	Spatio-temporal prefetching	Correlation across adjacent layers + consecutive tokens for staging	Reduced cache misses	Expert-level; requires cross-layer history buffer	Column-level temporal prediction

[!WARNING] **Lossy vs. Lossless:** Papers #12, 27, 28, 29 all achieve lower latency by **degrading model quality** (dropping experts, reducing precision, or changing routing). AAEC v3 is designed as **strictly lossless** — every column that the true router selects is fetched and computed, either via the fast path (prefetched) or slow path (demand-loaded).

6. Category C4: Neuron-Level & Activation Sparsity

These papers establish the **empirical foundation** for AAEC v3's column-level decomposition.

#	Paper	Year	Idea	Method	Key Results	Weaknesses	Future Work
---	-------	------	------	--------	-------------	------------	-------------

31	PowerInfer (Song et al., SJTU)	2024	Hot/cold neuron classification	GPU preloads hot neurons, CPU computes cold neurons	11.69× speedup over llama.cpp on consumer GPU	Dense models only (not MoE); hot/cold split is static (offline profiling)	Dynamic hot/cold adaptation; MoE extension
32	PowerInfer-2 (2024)	2024	Neuron clusters on smartphone	Polymorphic engine: NPU (hot clusters) + CPU (cold clusters)	47B model on smartphone at 11.68 tok/s	Mobile-specific (NPU pipeline); still neuron-cluster level, not column-level	Server-side neuron-level serving
33	Deja Vu (Liu et al., ICML 2023)	2023	Contextual sparsity prediction	Lightweight MLP predictors predict active neurons/heads per input	2× wall-clock speedup on OPT-175B	Dense models only; requires training sparsity predictors; no caching/offloading system	Integration with offloading systems
34	MoEfication (Zhang et al.)	2022	Convert dense FFN -> MoE	Partition FFN weights into expert groups via activation clustering	Simulates MoE-like sparsity	Requires retraining/fine-tuning; changes model structure	Training-free conversion
35	TEAL (2024)	2024	Training-free activation sparsity	Magnitude-based thresholding on hidden states at runtime	Plug-and-play 2× speedup	Works with ReLU; limited effectiveness with SiLU/SwiGLU (as used in Qwen3)	Smooth activation support
36	R-Sparse (2024)	2024	ReLU reintroduction for sparsity	Replace SwiGLU -> ReLU to recover natural sparsity	Up to 50% neuron pruning at runtime	Requires fine-tuning; changes activation function (lossy if not retrained)	Activation-preserving sparsity
37	FANG (2025)	2025	Function-aware neuron grouping	Calibration-based grouping of neurons into shared vs. routed sets	Better pruning accuracy	Structured pruning (permanent removal), not dynamic serving	Dynamic FANG for serving

*[!NOTE] Key observation: PowerInfer (paper #31) is the closest system to AAEC v3's neuron-level philosophy, but it targets **dense models only**. No existing work combines neuron/column-level granularity with MoE expert offloading. This is AAEC v3's core novelty.*

7. Category C5: Kernel Fusion & GPU Optimization

#	Paper	Year	Idea	Method	Key Results	Weaknesses
38	Triton Fused MoE (vLLM)	2024	Fused MoE dispatch kernel	Single Triton kernel handles routing, permutation, GEMM, combination	Standard in vLLM/TGI	Operates on full experts; no gather-scatter for sub-expert columns
39	Column-Major GEMM Scheduling (PyTorch)	2024	Locality-aware GEMM	Column-major work decomposition for L2 cache efficiency	4× speedup on H100/A100	Standard dense GEMM; not adapted for sparse column-indexed gather

40	FlashAttention-3 (Dao, 2024)	2024	Asynchronous pipelining in attention	Overlap GEMM, softmax, and memory ops in attention kernel	Near hardware-peak throughput	Attention-only optimization; does not address FFN/expert weight loading
----	--	------	--	--	-------------------------------------	---

8. Category C6: LLM Serving Infrastructure

#	Paper	Year	Idea	Method	Key Results	Weaknesses
41	vLLM / PagedAttention (Kwon et al.)	2023	Virtual memory paging for KV cache	Block-based KV allocation with copy-on-write	2-4× throughput over Orca	KV-cache only; does not manage expert weights
42	FlexGen (Sheng et al.)	2023	Offloading for throughput	Linear programming for GPU-CPU-disk weight placement	High throughput on single GPU	Throughput-oriented; high latency per token; not latency-sensitive

9. Category C7: Model Compression (Orthogonal)

#	Paper	Year	Idea	Method	Weakness for Our Context
43	GPTQ (Frantar et al.)	2023	Post-training quantization	Hessian-based error compensation	Lossy; perplexity degradation at low bits
44	AWQ (Lin et al.)	2024	Activation-aware quantization	Protect salient weights based on activation	Complementary to AAEC v3, not competing
45	SqueezeLLM (Kim et al.)	2024	Dense+Sparse decomposition	Non-uniform quantization + sparse outlier storage	Orthogonal; can be combined with column-level caching

10. Cross-Paper Gap Analysis

Repeated Limitations (Found in ≥3 Papers)

Limitation	Papers Affected	How AAEC v3 Addresses It
Expert-level granularity (9.44 MB transfer units)	#9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23 (ALL C2)	Column-level granularity (30.72 KB transfer units) — 307× finer
Inter-layer prediction (predict from previous layer output)	#9, 10, 22, 24	Intra-layer pre-attention prediction — uses same-layer pre-attention hidden state
No attention-window hiding (transfer happens after routing)	#9, 10, 11, 13, 14, 15, 16, 21	Shifts prefetch trigger to pre-attention; hides transfer behind attention compute window
Lossy serving (drops experts, changes precision, alters routing)	#12, 27, 28, 29	100% lossless — all missed columns demand-loaded before execution
LRU-only eviction (no lookahead)	#9, 10, 14, 16	Least-Stale eviction with Markov-based lookahead prediction
No kernel fusion for sub-expert columns	ALL C5 papers	Triton Gather-GEMM — custom fused kernel for indexed column access

Contradictory Results

Contradiction	Papers	Resolution
"Speculative prefetching solves the problem" vs. "Speculation wastes bandwidth when wrong"	CommitMoE (#12) vs. MoE-Infinity (#10)	AAEC v3 resolves via confidence-gated speculation : speculate when confident, demand-load when uncertain
"LRU is sufficient for expert caching" vs. "LRU fails under domain shifts"	Mixtral-Offloading (#9) vs. ADEPT (#13)	AAEC v3: Least-Stale with dynamic LRU-HP fallback under high routing entropy
"Neuron-level sparsity only works with ReLU" vs. "SwiGLU has activation concentration"	TEAL (#35) vs. R-Sparse (#36)	AAEC v3 empirically demonstrates 50% energy concentration in 115.5/768 columns with SwiGLU on Qwen3

Open Research Questions (From "Future Work" Sections)

Open Question	First Stated By	Status in AAEC v3
"Sub-expert offloading"	MoE-Infinity (#10), EdgeMoE (#16), ProMoE (#22)	■ Solved : Column-level (30.72 KB) granularity
"Intra-layer prediction (not inter-layer)"	Pre-gated MoE (#24)	■ Solved : Pre-attention router at same-layer input
"Lossless speculative serving"	MoE-SpeQ (#26), SiDA-MoE (#29)	■ Solved : Demand-load fallback guarantees 0% degradation
"Dynamic hot/cold adaptation"	PowerInfer (#31)	■ Solved : Virtual page remapping + entropy-aware eviction
"Server-side neuron-level serving"	PowerInfer-2 (#32)	■ Solved : Column-level serving on GPU servers
"Integration of sparsity prediction with offloading"	Deja Vu (#33)	■ Solved : Energy-ranked columns guide prefetch priority
"Latency-oriented MoE serving"	MoE-Lightning (#11), MoE-Gen (#20)	■ Addressed : Sub-millisecond stall target via attention-window hiding
"Column-level temporal prediction"	ST-MoE (#30)	■ Solved : Markov transition matrix for sequential column access

11. Novelty Positioning: What AAEC v3 Does That Nobody Else Does

What Has Been Done

Capability	Existing Work
Expert-level offloading & caching	Mixtral-Offloading, MoE-Infinity, DALI, EdgeMoE, + 15 others
Expert-level speculative prefetching	Pre-gated MoE, CommitMoE, ProMoE, SpecMoE
Neuron-level sparsity exploitation	PowerInfer, Deja Vu, TEAL (dense models only)
LRU/LFU expert cache eviction	Almost all C2 papers
Triton kernel fusion for MoE	vLLM Fused MoE, PyTorch column-major GEMM
Lossy serving for latency reduction	CommitMoE, AdapMoE, HOBBIT, SiDA-MoE

What Has NOT Been Done (AAEC v3 Novelty)

Novel Contribution	Why It's New	Closest Prior Work	Difference
Column-level MoE serving	All prior MoE serving operates at expert-level (9.44 MB). AAEC v3 operates at column-level (30.72 KB) — the first sub-expert decomposition for MoE offloading	PowerInfer (dense models)	PowerInfer targets dense FFNs with ReLU; AAEC v3 targets MoE experts with SwiGLU, using energy-ranked columns
Intra-layer pre-attention prefetching	All prior speculative prefetching predicts from the previous layer or token. AAEC v3 predicts from the same layer's pre-attention state, hiding transfers behind the same layer's attention window	Pre-gated MoE (inter-layer)	Pre-gated MoE uses layer L-1 output; AAEC v3 uses layer L input (pre-attention state {h}_{pre})
Energy-based column priority scheduling	No prior work ranks individual weight columns by activation energy to determine prefetch priority	Deja Vu (contextual sparsity)	Deja Vu predicts binary active/inactive neurons; AAEC v3 ranks columns by continuous energy magnitude for priority queuing
Least-Stale eviction with LRU-HP fallback	No prior cache eviction policy uses Markov-based sequential lookahead with entropy-gated dynamic fallback	LRU (all C2 papers)	LRU evicts by recency; LS evicts by predicted next-access distance; falls back to LRU-HP under high entropy
Lossless attention-window-hidden serving	No prior system achieves 0% accuracy degradation while hiding 100% of weight transfers behind the attention compute window	CommitMoE (low stall but lossy)	CommitMoE drops 7/8 experts; AAEC v3 fetches all via fast/slow paths
Triton Gather-GEMM for indexed columns	No existing kernel fuses indexed column gathering with GEMM and pointwise accumulation in a single pass	vLLM Fused MoE (full expert)	vLLM kernel processes full expert matrices; AAEC v3 kernel gathers arbitrary column subsets from multiple sources
Dual-partition ADETR with virtual page remapping	No prior MoE serving system uses virtual-to-physical page table remapping to maintain contiguous memory layouts without physical copies	PagedAttention (KV cache)	PagedAttention pages KV cache; ADETR pages expert weight columns with zero-copy remapping

12. Summary: Why AAEC v3 Is New

The literature reveals a clear **granularity wall**: all MoE serving systems operate at expert-level. Meanwhile, dense-model research (PowerInfer, Deja Vu) has shown that neuron/column-level sparsity is highly exploitable — but only for dense architectures with ReLU activations.

AAEC v3 bridges this gap. It is the first system to:

- 1. **Decompose MoE expert FFN matrices into individual column packets** (30.72 KB) and cache/prefetch at that granularity
- 2. **Exploit within-expert activation energy concentration** (50% energy in 15% of columns) even with SwiGLU activations
- 3. **Hide weight transfers behind the attention compute window** of the same layer (not the previous layer)
- 4. **Guarantee 100% lossless serving** with a binary speculate/demand-load policy

No existing paper in the surveyed 45 works combines all four of these properties.

Prior Art Design Space:		
+-----+		
Expert-Level -----	Neuron-Level	
(MoE Serving)	(Dense Models)	
* CommitMoE	* PowerInfer	
* MoE-Infinity	* Deja Vu	
* DALI	* TEAL	
* EdgeMoE		
* ProMoE		
+=====+		
AAEC v3	<- Column-Level	
(This Work)	MoE Serving	
+=====+		
	(NEW)	
Lossy -----	Lossless	
* CommitMoE	* AAEC v3	
* AdapMoE	* Mixtral-Offloading	
* HOBBIT	* MoE-Infinity	
* SiDA-MoE		
+-----+		